CHAPTER EXERCISE: Answers for this chapter start on page 296.

A calculator should NOT be used on the following questions.

1

$$3x - 5y = -11$$
$$x = 1 - 3y$$

What is the solution (x, y) to the system of equations above?

A) $(-5, 2)$
B) $(-2, 1)$
C) $(1, 0)$
D) $(4, -1)$

2

$$y + 2x = 20$$
$$6x - 5y = 12$$

What is the solution (x, y) to the system of equations above?

A) $(-7, 6)$
B) $(-6, 7)$
C) $(6, 7)$
D) $(7, 6)$

3

$$3x - 4y = 21$$
$$4x - 3y = 14$$

If (x, y) is a solution to the system of equations above, what is the value of $y - x$?

A) -18
B) -5
C) 5
D) 8

4

$$2x + 5y = 24$$
$$x + 4y = 15$$

If (x, y) satisfies the system of equations above, what is the value of $x + y$?

A) 7
B) 8
C) 9
D) 10

5

$$3x + y = -2x + 8$$
$$-3x + 2y = -10$$

If (x, y) is a solution to the system of equations above, what is the value of xy ?

A) -16
B) -8
C) -4
D) 4

6

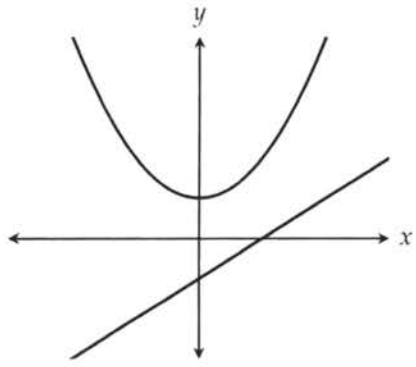
$$y = ax + b$$
$$y = -bx$$

The equations of two lines in the xy-plane are shown above, where a and b are constants. If the two lines intersect at $(2, 8)$, what is the value of a ?

A) 2
B) 4
C) 6
D) 8

7

$$y = x^2 + 1$$
$$y = x - 1$$



A system of two equations and their graphs in the xy-plane are shown above. How many solutions does the system have?

A) Zero
B) One
C) Two
D) Three

8

$$-5x = y + 2$$
$$2(2x - 1) = 3 - 3y$$

What is the solution (x, y) to the system of equations above?

A) $(-2, 8)$
B) $(-1, 3)$
C) $(1, -7)$
D) $(3, -17)$

9

$$2x - 4y = 8$$
$$x + 2y = 4$$

How many solutions (x, y) are there to the system of equations above?

A) Zero
B) One
C) Two
D) More than two

10

$$2x - 5y = a$$
$$bx + 10y = -8$$

In the system of equations above, a and b are constants. If the system has infinitely many solutions, what is the value of a ?

A) -4
B) $\frac{1}{4}$
C) 4
D) 16

11

$$ax + 2y = 5$$
$$3x - 6y = 20$$

In the system of equations above, a is a constant. If the system has one solution, which of the following can NOT be the value of a ?

A) -1
B) $\frac{3}{4}$
C) 1
D) 3

12

$$4x - \frac{1}{3}y = -8$$

$$y = 4x + 16$$

What is the solution (x, y) to the system of equations above?

A) $(-2, 8)$
B) $(-1, 12)$
C) $(1, 20)$
D) $(3, 28)$

13

$$y = 0.5x + 14$$

$$x - y = -18$$

According to the system of equations above, what is the value of y ?

14

$$\frac{1}{3}x - \frac{1}{6}y = 4$$

$$6x - ay = 8$$

In the system of equations above, a is a constant. If the system has no solution, what is the value of a ?

A) $\frac{1}{3}$
B) 1
C) 3
D) 6

15

$$3x - 6y = 15$$

$$-2x + 4y = -10$$

How many solutions (x, y) are there to the system of equations above?

A) Zero
B) One
C) Two
D) More than two

16

$$mx - 6y = 10$$

$$2x - ny = 5$$

In the system of equations above, m and n are constants. If the system has infinitely many solutions, what is the value of $\frac{m}{n}$?

A) $\frac{1}{12}$
B) $\frac{1}{3}$
C) $\frac{4}{3}$
D) 3

17

$$y = \sqrt{x} + 3$$

$$\sqrt{4x} - y = 3$$

If (x, y) is the solution to the system of equations above, what is the value of y ?

A calculator is allowed on the following questions.

18

A local supermarket sells jelly in small, medium, and large jars. Sixteen small jars weigh as much as two medium jars and one large jar. Four small jars and one medium jar have the same weight as one large jar. How many small jars have the weight of one large jar?

A) 7
B) 8
C) 9
D) 10

19

On a math test with 30 questions, 5 points are rewarded for each correct answer and 2 points are deducted for each incorrect answer. If James answered all the questions and scored 59 points, solving which of the following systems of equations gives his number of correct answers, x, and his number of incorrect answers, y, on the math test?

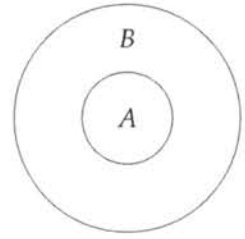
A) $x + y = 59$
$5x - 2y = 30$

B) $x + y = 30$
$5x + 2y = 59$

C) $x + y = 30$
$2x - 5y = 59$

D) $x + y = 30$
$5x - 2y = 59$

20



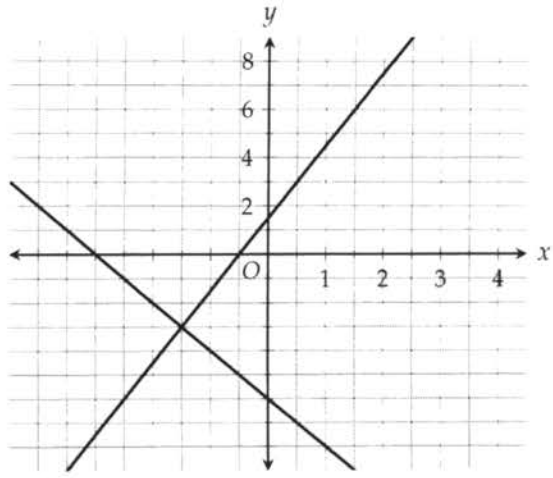
A game of darts rewards points depending on which region is hit. There are two regions, A and B, as shown above. James throws 3 darts, hitting region A once and region B twice, for a total of 18 points. Oleg also throws 3 darts, but hits regions A twice and region B once for a total of 21 points. How many points are rewarded for hitting region B once?

21

A restaurant has two types of tables, rectangular ones that can each seat 4 people and circular tables that can each seat 8 people. If 144 people are enough to fill all 30 tables at the restaurant, how many rectangular tables does the restaurant have?

A) 12
B) 16
C) 20
D) 24

22



A system of two equations is graphed in the xy-plane above. Which of the following is the solution (x, y) to the system?

A) $(0, -6)$

B) $(-3, -3)$

C) $\left(-\frac{3}{2}, -3\right)$

D) $\left(-3, -\frac{5}{2}\right)$

23

$$x^2 - y^2 = \frac{1}{12}$$

$$x - 2y = 0$$

If the ordered pairs (x_1, y_1) and (x_2, y_2) satisfy the system of equations above, what are the values of y_1 and y_2 ?

A) $-\frac{1}{2}$ and $\frac{1}{2}$

B) $-\frac{1}{\sqrt{12}}$ and $\frac{1}{\sqrt{12}}$

C) $-\frac{1}{4}$ and $\frac{1}{4}$

D) $-\frac{1}{6}$ and $\frac{1}{6}$

24

In the xy-plane, the graph of $y = x^2 - 7x + 7$ intersects the graph of $y = 2x - 1$ at the points $(1, 1)$ and (p, q). What is the value of p ?

25

$$x^2 - 2x = y - 1$$

$$x = y - 11$$

If (x, y) is a solution to the system of equations above, what is one possible value of y ?

Chapter 10: Systems of Equations

CHAPTER EXERCISE:

1. [B] Substituting the second equation into the first,

$$\begin{aligned} 3(1-3y)-5y &= -11 \\ 3-9y-5y &= -11 \\ 3-14y &= -11 \\ -14y &= -14 \\ y &= 1 \end{aligned}$$

Finally, $x = 1-3(1) = -2$.

2. [D] From the first equation, $y = 20-2x$. Plugging this into the second equation,

$$\begin{aligned} 6x-5(20-2x) &= 12 \\ 6x-100+10x &= 12 \\ 16x &= 112 \\ x &= 7 \end{aligned}$$

We already know the answer is (D) at this point, but just in case, $y = 20-2(7) = 6$.

3. [B] Add the two equations to get $7x-7y = 35$. Dividing both sides by 7, $x-y = 5$. We can multiply both sides by -1 to get $y-x = -5$.

4. [C] The fastest way to do this problem is to subtract the second equation from the first, which yields $x+y = 9$.

5. [C] In the first equation, we can move $3x$ to the right hand side to get $y = -5x+8$. Substituting this into the second equation,

$$\begin{aligned} -3x+2(-5x+8) &= -10 \\ -3x-10x+16 &= -10 \\ -13x &= -26 \\ x &= 2 \end{aligned}$$

Then, $y = -5(2)+8 = -2$. Finally, $xy = (2)(-2) = -4$.

6. [C] If the two lines intersect at the point $(2,8)$, then $(2,8)$ is a solution to the system. Plugging the point into the equation of the second line, we can solve for b,

$$\begin{aligned} y &= -bx \\ 8 &= -b(2) \\ -4 &= b \end{aligned}$$

Plugging the point into the equation of the first line,

$$\begin{aligned} y &= ax+b \\ 8 &= a(2)-4 \\ 12 &= 2a \\ 6 &= a \end{aligned}$$

7. [A] The two graphs do not intersect at all, so there are no solutions.

8. [B] From the first equation, we can isolate y to get $y = -5x-2$. Substituting this into the second equation,

$$\begin{aligned} 2(2x-1) &= 3-3(-5x-2) \\ 4x-2 &= 3+15x+6 \\ 4x-2 &= 15x+9 \\ -11 &= 11x \\ -1 &= x \end{aligned}$$

Finally, $y = -5(-1)-2 = 3$.

9. [B] Divide the first equation by 2 to get $x-2y = 4$. We can't get the coefficients to match (-2 vs. 2 for the y's). Therefore, the system has one solution. In fact, we can even solve this system by adding the two equations to get $2x = 8$, $x = 4$, which makes $y = 0$.

10. [C] To get the same coefficients, multiply the first equation by -2 to get $-4x+10y = -2a$. Now we can see that $-2a = -8$, $a = 4$.

11. [A] Multiply the first equation by -3 to get $-3ax - 6y = -15$. The constant a cannot be -1. Otherwise, the second equation's coefficients would then be equal to the first equation's coefficients, resulting in a system with no solution.

12. [B] First, multiply the first equation by 3 to get rid of the fraction: $12x - y = -24$. Next, substitute the second equation into the first,

$$\begin{aligned} 12x - (4x + 16) &= -24 \\ 12x - 4x - 16 &= -24 \\ 8x &= -8 \\ x &= -1 \end{aligned}$$

Finally, $y = 4(-1) + 16 = 12$.

13. [10] We can isolate x in the second equation to get $x = y - 18$. Substituting this into the first equation,

$$\begin{aligned} y &= 0.5(y - 18) + 14 \\ y &= 0.5y - 9 + 14 \\ 0.5y &= 5 \\ y &= 10 \end{aligned}$$

14. [C] To match the coefficients, multiply the first equation by 18 to get $6x - 3y = 72$. We can then see that $a = 3$ if the system is to have no solution.

15. [D] Divide the first equation by 3 to get $x - 2y = 5$. Divide the second equation by -2 to get $x - 2y = 5$. They're the same, so there are an infinite number of solutions.

16. [C] For a system to have infinitely many solutions, the equations must essentially be the same. Looking at the constants, we can make them match by multiplying the second equation by 2. The equations then look like this:

$$\begin{aligned} mx - 6y &= 10 \\ 4x - 2ny &= 10 \end{aligned}$$

Now it's easy to see that $m = 4$ and $2n = 6$, $n = 3$. Finally, $\frac{m}{n} = \frac{4}{3}$.

17. [9] Plugging the first equation into the second,

$$\begin{aligned} \sqrt{4x} - (\sqrt{x} + 3) &= 3 \\ 2\sqrt{x} - \sqrt{x} - 3 &= 3 \\ \sqrt{x} &= 6 \\ x &= 36 \end{aligned}$$

Therefore, $y = \sqrt{36} + 3 = 9$.

18. [B] Let s, m, and l be the weights of small, medium, and large jars, respectively. Based on the information, we can create the following two equations:

$$\begin{aligned} 16s &= 2m + l \\ 4s + m &= l \end{aligned}$$

To get the weight of the large jar in terms of the weight of the small jar, we need to get rid of m, the weight of the medium jar. We could certainly use elimination, but here, we'll use substitution. Isolating m in the second equation, $m = l - 4s$. Substituting this into the first equation, we get

$$\begin{aligned} 16s &= 2(l - 4s) + l \\ 16s &= 2l - 8s + l \\ 24s &= 3l \\ 8s &= l \end{aligned}$$

Eight small jars are needed to match the weight of one large jar.

19. [D] Since there were 30 questions, James must have had 30 answers, $x + y = 30$. The points he earned from correct answers total $5x$. The points he lost from incorrect answers total $2y$. Therefore, $5x - 2y = 59$.

20. [5] Let a and b be the number points you get for hitting regions A and B, respectively. From the information, we can form the following two equations:

$$\begin{aligned} a + 2b &= 18 \\ 2a + b &= 21 \end{aligned}$$

To solve for b, multiply the first equation by 2 and subtract to get $3b = 15$, $b = 5$.

21. [D] Let r and c be the number of rectangular tables and circular tables, respectively, at the restaurant. Based on the information, we can make the following two equations:

$$4r + 8c = 144$$
$$r + c = 30$$

To solve for r, multiply the second equation by 8 and subtract to get $-4r = -96$, $r = 24$.

22. [C] The solution to the system is the intersection point of the two lines. Each horizontal step along the grid represents $\frac{1}{2}$ of a unit, and each vertical step along the grid represents 1 unit. So, the intersection point is at $\left(-\frac{3}{2}, -3\right)$.

23. [D] From the second equation, $x = 2y$. Plugging this into the first equation, we get

$$(2y)^2 - y^2 = \frac{1}{12}$$
$$4y^2 - y^2 = \frac{1}{12}$$
$$3y^2 = \frac{1}{12}$$
$$y^2 = \frac{1}{36}$$
$$y = \pm\frac{1}{6}$$

Therefore, the values of y_1 and y_2 are $-\frac{1}{6}$ and $\frac{1}{6}$.

24. [8] To find the point(s) where two graphs intersect, solve the system consisting of their equations. In this problem, that system is

$$y = x^2 - 7x + 7$$
$$y = 2x - 1$$

Substituting the first equation into the second, we get

$$x^2 - 7x + 7 = 2x - 1$$
$$x^2 - 9x + 8 = 0$$
$$(x-1)(x-8) = 0$$
$$x = 1 \text{ or } 8$$

So the x-coordinates of the points of intersection are 1 and 8. Since the question already gave us the point $(1, 1)$, p must be equal to 8.

25. [9 or 16] First, add 11 to both sides of the second equation to get $y = x + 11$. Then substitute this in for y in the first equation:

$$x^2 - 2x = x + 11 - 1$$
$$x^2 - 3x - 10 = 0$$
$$(x+2)(x-5) = 0$$
$$x = -2 \text{ or } 5$$

When $x = -2$, $y = -2 + 11 = 9$. When $x = 5$, $y = 5 + 11 = 16$. The solutions to the system are then $(-2, 9)$ and $(5, 16)$. Therefore, the possible values of y are 9 and 16.